MIL-C-24231/22D(SH)
15 May 1987
SUPERSEDING
MIL-C-24231/22C(SHIPS)
30 August 1976
(See note 4)

MILITARY SPECIFICATION SHEET

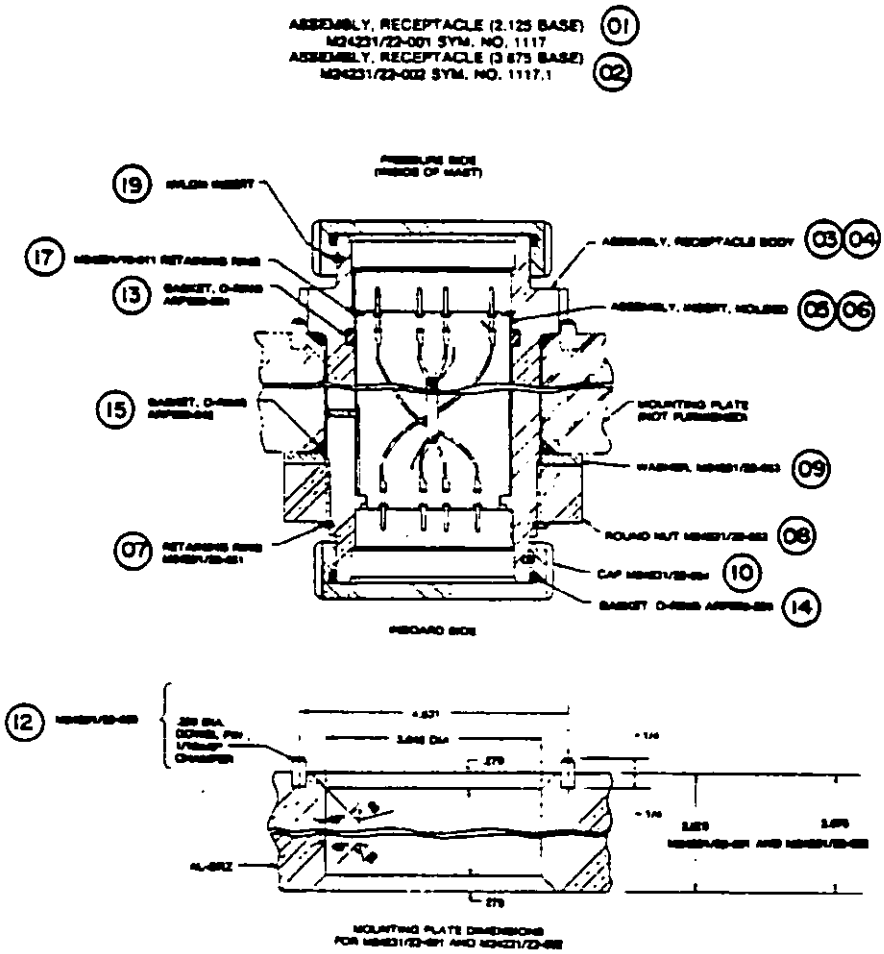
CONNECTOR, RECEPTACLE, TYPE IV, 80-CONDUCTOR

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the receptacle described herein shall consist of this specification and the latest issue of MIL-C-24231.

AMSC N/A FSC 5935
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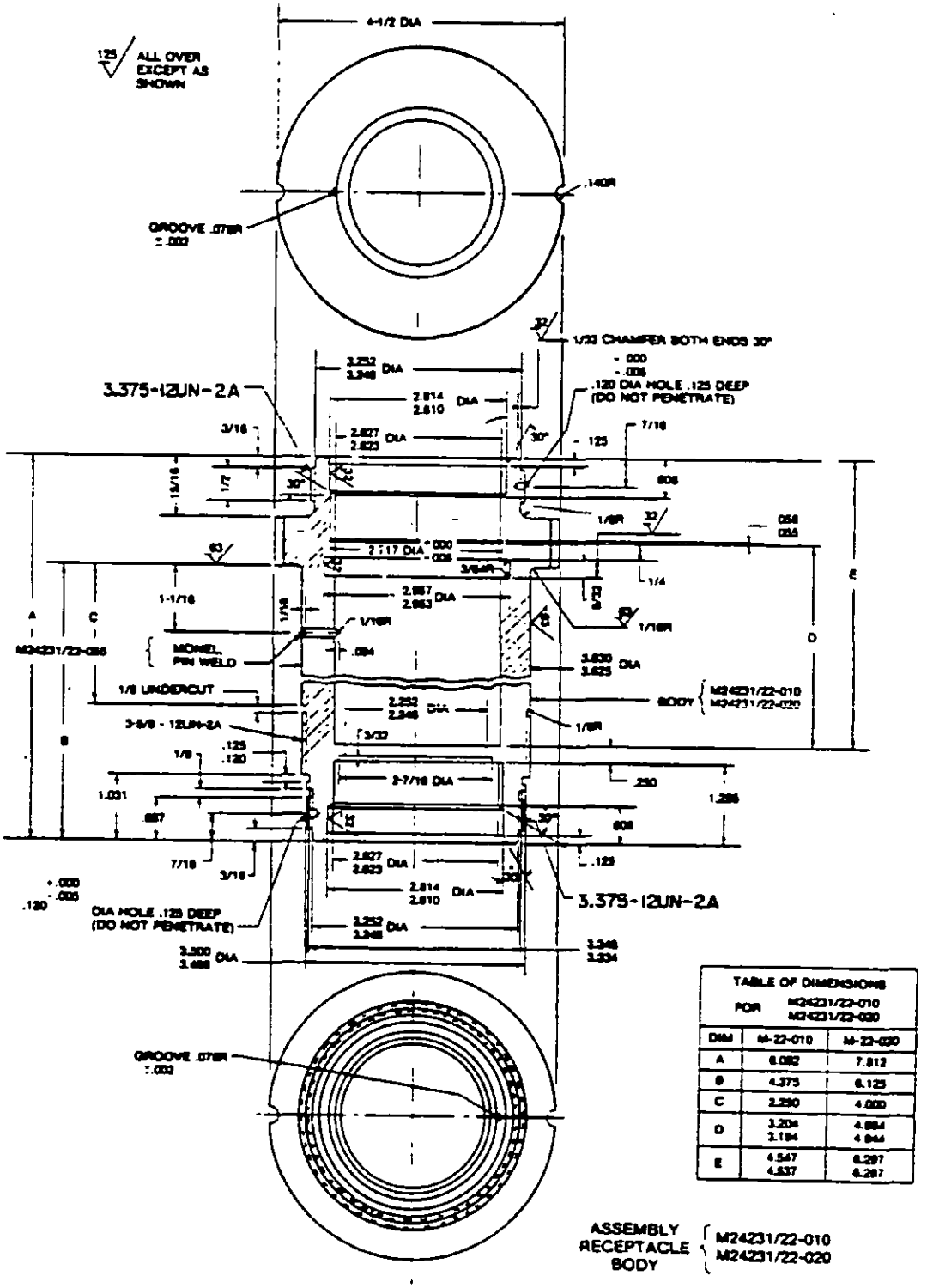
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SH 9689

FIGURE 1. Connector, receptacle, 80-conductor - typical.

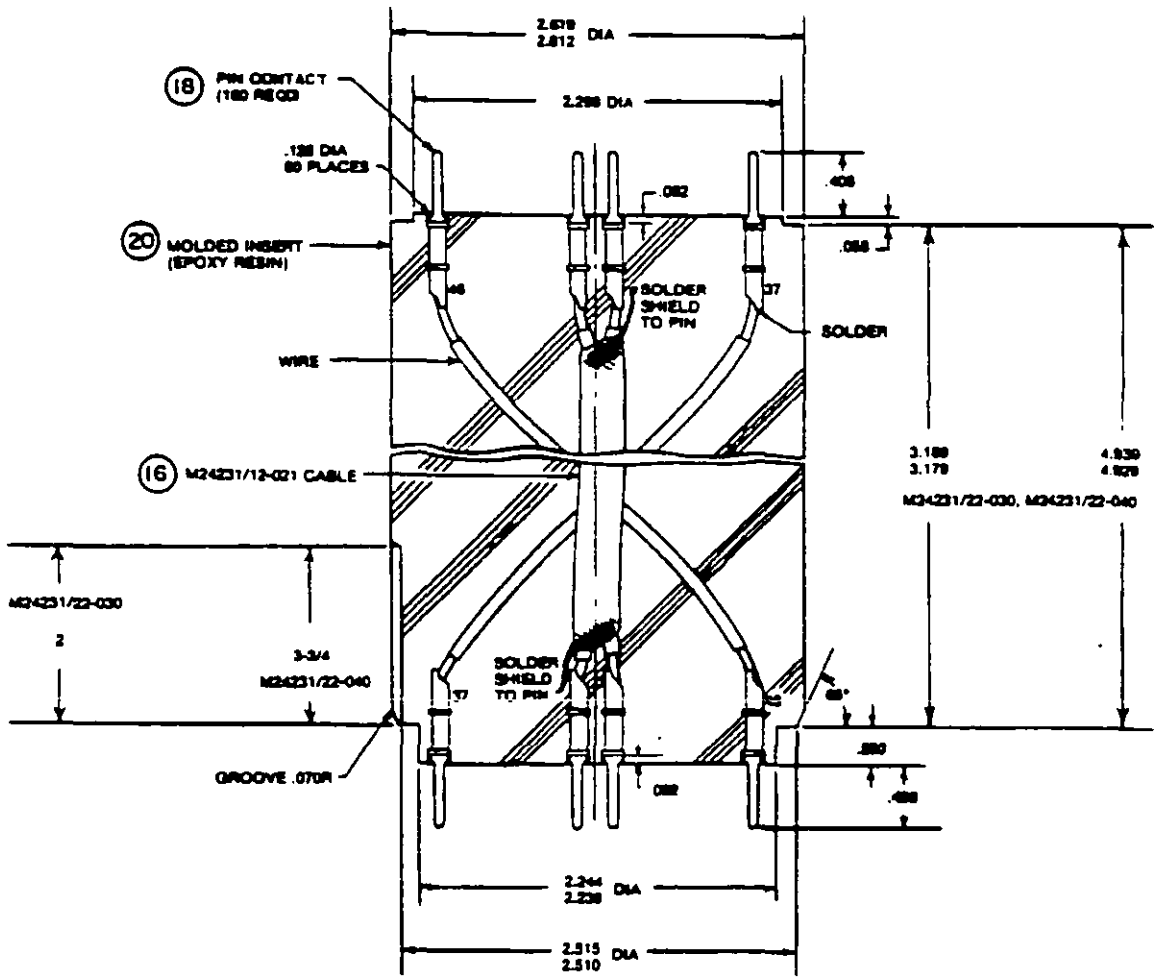
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SH 9689-1

FIGURE 2. Assembly, receptacle body. (03) (04)

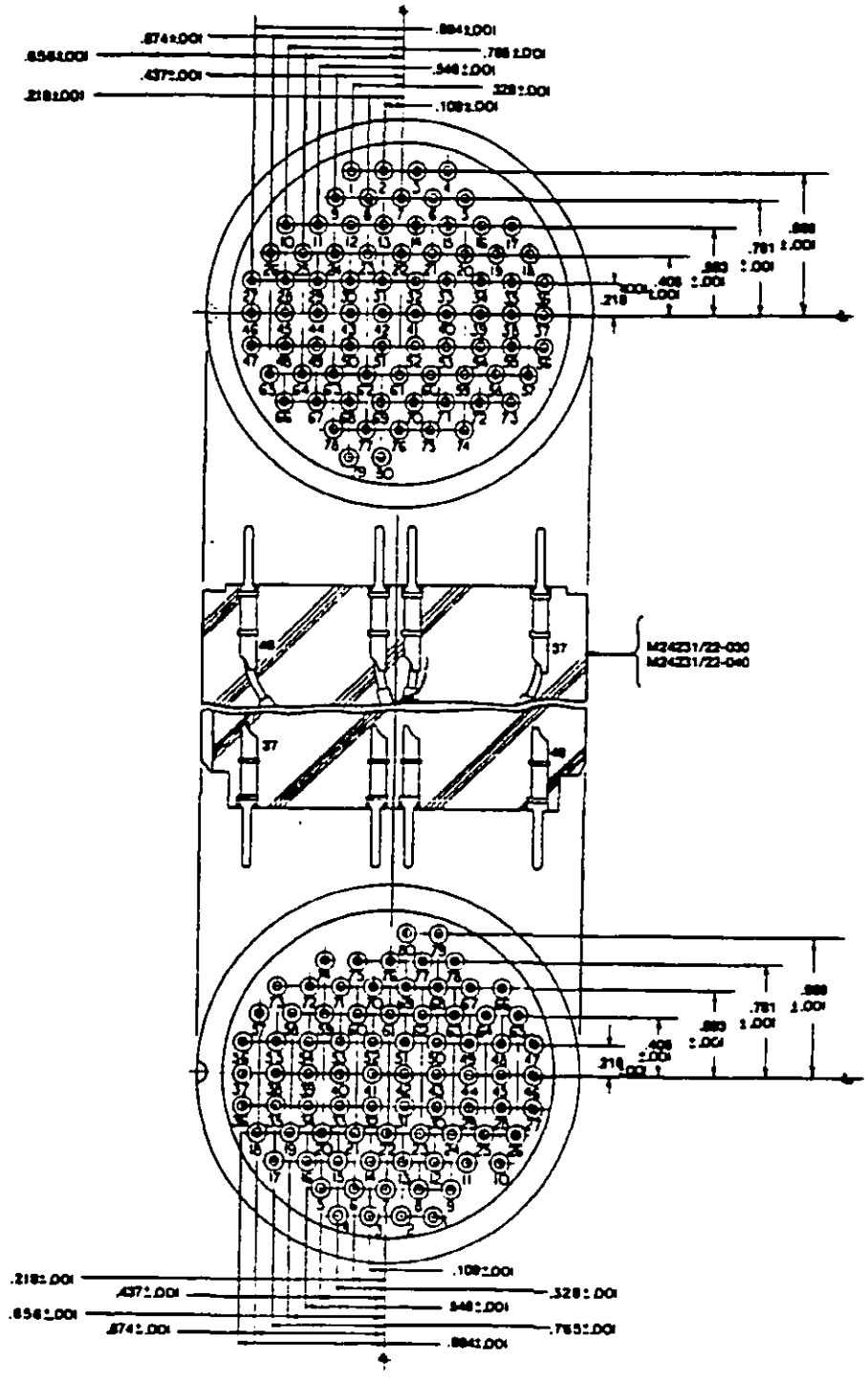
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SH 9689-2

FIGURE 3. Assembly, insert, molded - M24231/22-030, M24231/22-040. (05) (06)

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SH 9689-3

FIGURE 4. Insert, pin contact layout.

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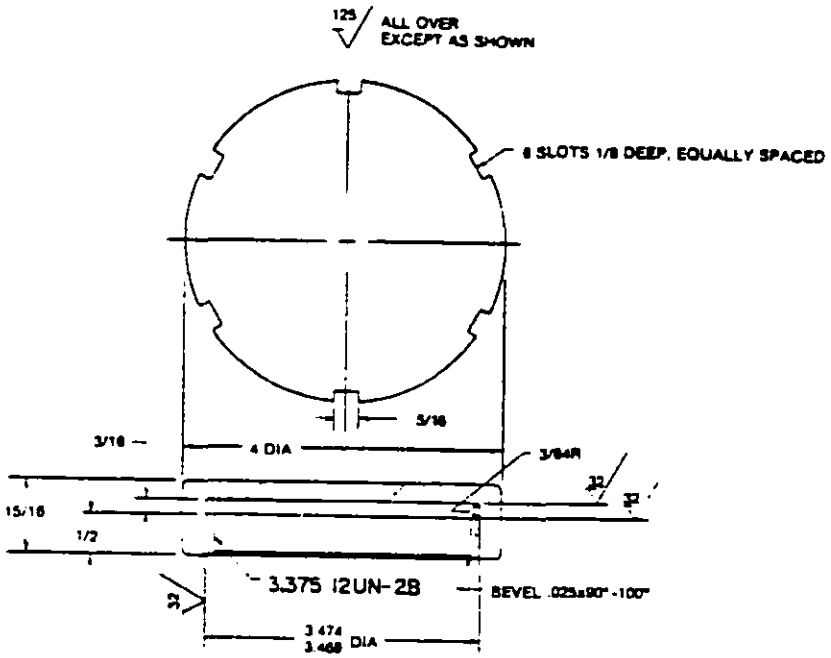
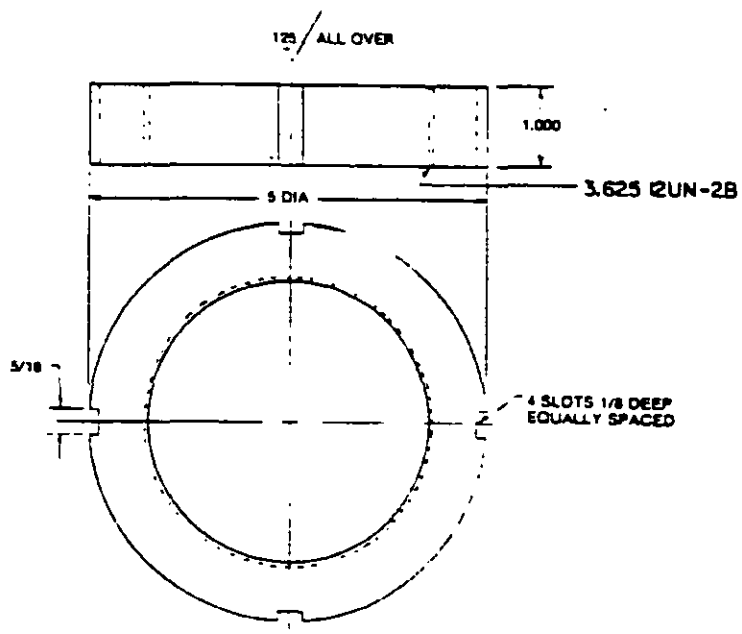


FIGURE 5. Cap - M24231/22-054. (10)



SH 9689-4

FIGURE 6. Nut - M24231/22-052.

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REQUIREMENTS:

1. Dimensions and configuration: See figures.
2. Each nut, washer, and receptacle body shall be examined for cracks in accordance with the liquid penetrant method of MIL-STD-271. Finding of cracks shall be cause for rejection.
3. Welding of the pin is allowed after the hull fitting body has been age-hardened to achieve its required physical properties.

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TABLE I. Parts list.

Item 1/ no.	Description 2/	Quantity for M24231/22- 3/ 001	Quantity for M24231/22- 3/ 002	Material/specification	Remarks
01	Receptacle, 80-conductor, symbol no. 1117	1		-----	See figure 1
02	Receptacle, 80-conductor, symbol no. 1117.1		1	-----	See figure 1
03	Body, receptacle M24231/22-010	1		Ni-Cu-Al, QQ-N-286	See figure 2
04	Body, receptacle M24231/22-020		1	Ni-Cu-Al, QQ-N-286	See figure 2
05	Insert, assembly, molded	1		See items 16, 18, and 20	See figure 3
06	Insert assembly, molded		1	See items 16, 18, and 20	See figure 3
07	Ring, retaining	1	1	SPIROLOX RST-350-S 6/	See figure 1
08	Nut, round	1	1	Ni-Cu-Al, QQ-N-286	See figure 1
09	Washer, 3.640/3.635 by 5.187 by 0.187	1	1	Ni-Cu-Al, QQ-N-286	See figure 1
10	Cap, pressure-proof	2	2	Ni-Al-BRZ, QQ-C-465 5/	See figure 5
11	Pin, guide, 0.125 diameter, length as required	1	1	Ni-Cu, class A, QQ-N-281	See figure 2
12	Pin, dowel	2	2	Ni-Cu, class A, QQ-N-281	See figure 1
13	Gasket, O-ring	1	1	MS28775-334	Lubricate with silicone compound, MIL-S-8660, see figure 1

See footnotes at end of table.

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TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	Quantity for M24231/22- 3/ 001	Quantity for M24231/22- 3/ 002	Material/specification	Remarks
14	Gasket, O-ring	2	2	MS28775-236	See item 13
15	Gasket, O-ring	2	2	MS28775-342	See item 13
16	Cable, 2-conductor, shielded, 2SWU-1, length as required	8	8	MIL-C-24643/33	See figure 3
17	Ring, retaining	1	1	M24231/19-011	See figure 1
18	Pin, contact	160	160	MIL-C-24231	See figure 3
19	Insert, 0.120 diameter, length as required	2	2	Nylon, group 1, class 1, ASTM D 4066	See figure 1
20	Epoxy resin, molded	As required		Cast epoxy resin compound 4/	See figure 3

1/ Item numbers are the same as encircled numbers shown on figures.
2/ See applicable figure or table listed in remarks column for part numbers and dimensions.
3/ Quantity is for one complete assembly of each size.
4/ Arcoset No. 200, M.M. Reif and Co., Rockledge, PA, or Phillicast No. 2, Philadelphia Resin Corp., Montgomeryville, PA, or equal.
5/ May be high impact plastic, PVC, coml at the option of the manufacturer.
6/ Manufactured by Ramsey Corp., St. Louis, MO, or equal.

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NOTES:

1. All dimensions are in inches.
2. Unless otherwise indicated, tolerances shall be as follows:

   (a) Decimals - plus or minus 0.005 inch.
   (b) Fractions - plus or minus 1/64 inch.
   (c) Angular dimensions - plus or minus 1/2 degree.

3. These connectors are for maintenance purposes only.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5935-N279-22)

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